

Generation Interconnection Request System Impact Study Report

For Queue Project Q23-2349

Elwood 230 kV
February 26, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2349. The Interconnection Customer proposes a 430.1 MW battery energy storage facility interconnecting at the Elwood 230 kV substation in the Illinois area. The request was received on September 14, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 487 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,766,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Bluebonnet – Russell Fork 230 KV
Contingency: Dekalb – Jones Chapel 500 KV
Loading: 102.9% (Rating: 2,526 MVA)
DFAX: 31.3%
Pre-project loading: 97.6%
2. Facility: Darlington Rd – Muskogee 100 KV
Contingency: Richland – Dekalb 500 KV L
Loading: 102.3% (Rating: 783 MVA)
DFAX: 6.5%
Pre-project loading: 98.7%
3. Facility: Ligonier Substation – Cypress Creek 100 KV
Contingency: Dekalb – Jones Chapel 500 KV L
Loading: 102.1% (Rating: 432 MVA)
DFAX: 5.2%
Pre-project loading: 96.9%
4. Facility: Robison Park – Russell Fork 230 KV
Contingency: Dekalb – Jones Chapel 500 KV
Loading: 100.6% (Rating: 3,415 MVA)
DFAX: 42.6%
Pre-project loading: 95.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Russell Fork – Bluebonnet 230 KV	Reconductor line	\$28,391,000	4% (\$1,009,000)
Muskogee – Darlington Road 100 KV	Reconductor line	\$7,317,000	4% (\$261,000)
Ligonier Substation – Cypress Creek 100 KV	Reconductor line	\$2,164,000	4% (\$75,000)
Robison Park – Russell Fork 230 KV	Reconductor line	\$11,853,000	4% (\$421,000)
	Total allocated to project		\$1,766,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.